

Linear Equations in One Variable

Concise Learning and Practice Guide

1. What does solving mean?

E1: A taxi charges a fixed fee of \$4 plus \$2 for each kilometre travelled. Suppose the total fare is \$18. **How many kilometres did the taxi travel?**

Let k = number of kilometres travelled. We can write $4 + 2k = 18$. To **solve** the equation means to find the value of k that makes the statement true:

$$4 + 2(7) = 18 \quad \Rightarrow \quad k = 7$$

The taxi travelled **7 km**.

E2: Two phone plans have different costs. Plan A charges \$10 plus \$3 per month, while Plan B charges \$22 plus \$2 per month. **After how many months will both plans have the same total cost?**

Let m = number of months. We can write $10 + 3m = 22 + 2m$. Solving gives

$$3m - 2m = 22 - 10 \quad \Rightarrow \quad m = 12$$

After **12 months**, both plans have the same total cost.

2. Basic solving process

Equations such as $ax + b = c$ and $ax + b = cx + d$ can be solved by:

1. Simplifying each side (distribute and combine like terms);
2. Collecting variable terms on one side;
3. Collecting constants on the other side;
4. Isolating the variable; and
5. Checking by substitution in the *original* equation.

Model: $3x + 5 = 20$

$$3x + 5 - 5 = 20 - 5 \quad (\text{subtract } 5)$$

$$3x = 15$$

$$\frac{3x}{3} = \frac{15}{3} \quad (\text{divide by } 3)$$

$$x = 5$$

Check: $3(5) + 5 = 20$, so the left side equals the right side. Thus $x = 5$.

Model: $5x + 4 = 2x + 19$

$$5x - 2x + 4 = 2x - 2x + 19 \quad (\text{subtract } 2x)$$

$$3x + 4 = 19$$

$$3x + 4 - 4 = 19 - 4 \quad (\text{subtract } 4)$$

$$3x = 15$$

$$\frac{3x}{3} = \frac{15}{3} \quad (\text{divide by } 3)$$

$$x = 5$$

Check: $5(5) + 4 = 29$ and $2(5) + 19 = 29$, so both sides are equal. Thus $x = 5$.

3. Worked examples

A. Form $ax + b = c$: $-4x - 7 = 17$

$$-4x - 7 + 7 = 17 + 7 \quad (\text{add } 7)$$

$$-4x = 24$$

$$\frac{-4x}{-4} = \frac{24}{-4} \quad (\text{divide by } -4)$$

$$x = -6$$

Check: $-4(-6) - 7 = 17$.

B. Form $ax + b = cx + d$: $-5x + 4 = 2x - 17$

$$-5x - 2x + 4 = 2x - 2x - 17 \quad (\text{subtract } 2x)$$

$$-7x + 4 = -17$$

$$-7x + 4 - 4 = -17 - 4 \quad (\text{subtract } 4)$$

$$-7x = -21$$

$$\frac{-7x}{-7} = \frac{-21}{-7} \quad (\text{divide by } -7)$$

$$x = 3$$

Check: $-5(3) + 4 = -11 = 2(3) - 17$.

C. Form $a(x + b) = c$: $-3(x - 2) = 18$

$$-3x + 6 = 18 \quad (\text{distribute } -3)$$

$$-3x + 6 - 6 = 18 - 6 \quad (\text{subtract } 6)$$

$$-3x = 12$$

$$\frac{-3x}{-3} = \frac{12}{-3} \quad (\text{divide by } -3)$$

$$x = -4$$

Check: $-3(-4 - 2) = 18$.

D. Form $\frac{x}{a} + b = c$: $-\frac{x}{4} + 3 = 8$

$$-\frac{x}{4} + 3 - 3 = 8 - 3 \quad (\text{subtract } 3)$$

$$-\frac{x}{4} = 5$$

$$(-4) \left(-\frac{x}{4} \right) = (-4)(5) \quad (\text{multiply by } -4)$$

$$x = -20$$

Check: $-\frac{-20}{4} + 3 = 8$.

4. Equations in context

Follow the chain

context $\longrightarrow$ equation $\longrightarrow$ solution $\longrightarrow$ interpretation.

Bike rental. A bike rental service charges a \$6 unlock-ing fee and \$3 per hour. The total cost is \$24. Let h be the number of hours.

$$6 + 3h = 24 \implies 3h = 18 \implies h = 6$$

The bike was rented for **6 hours**.

Online order. Four identical notebooks plus a \$3 deliv-ery charge cost \$27. Let p be the price of one notebook.

$$4p + 3 = 27 \implies 4p = 24 \implies p = 6$$

Each notebook costs **\$6**.

Saving. Zuleika already has \$25 and saves \$8 each week. When will she have \$81? Let w be the number of weeks.

$$25 + 8w = 81 \implies 8w = 56 \implies w = 7$$

Zuleika needs **7 weeks**.

6. Practice problems

For Problems 1–6, **define a variable, construct an equation, solve, and interpret** the answer.

- Two scooter companies offer different rental plans. Company A charges a \$6 unlocking fee plus \$4 per hour, while Company B charges a \$14 unlocking fee plus \$2 per hour. For how many hours is Company A cheaper than Company B?
- Two streaming services offer different plans. Service A charges \$3 per premium movie with no monthly fee, while Service B charges a \$12 monthly fee plus \$1 per premium movie. For how many premium movies is Service A the cheaper option?
- Cyprian is comparing two savings plans. Plan A starts with \$20 and adds \$8 each week, while Plan B starts with \$44 and adds \$5 each week. After how many weeks does Plan A contain more money than Plan B?
- A taxi company charges \$5 per kilometre with no initial fee. Another transportation service charges a flat fee of \$30 regardless of the distance travelled. For what trip distances is the taxi company the cheaper option?
- Two stores sell identical notebooks. Store A charges \$4 per notebook plus a \$10 delivery fee, while Store B charges \$6 per notebook with no delivery fee. For how many notebooks is Store A the cheaper option?
- Two parking garages charge according to the length of a stay. Garage A charges \$3 per hour plus a \$5 entrance fee, while Garage B also charges \$3 per hour but has a \$9 entrance fee. For what lengths of stay is Garage A cheaper than Garage B? Explain what your result means in context.

- Two bicycle rental shops use different pricing struc-tures. Shop A charges \$4 per hour plus a \$6 rental fee. Shop B charges a \$2 processing fee, a \$4 service fee, and \$4 per hour. For what rental durations is Shop A cheaper than Shop B? Explain what your result means in context.

For Problems 8–25, solve each equation.

- $x + 7 = 19$
- $5y = 35$
- $3z + 4 = 22$
- $4a - 7 = 2a + 9$
- $3(b + 2) = 21$
- $2(c + 5) + 3 = 19$
- $3(d + 2) + d = 18$
- $2(m + 4) + m = 2m + 17$
- $3(n + 2) + n = 2(n + 5) + 8$
- $5(p - 3) + 2p = 3(p + 1) + 10$
- $4(2q + 1) - 3q = 2(q + 7) + 5$
- $7 - 2(r + 4) = 3(r - 1) - 5$
- $3(2s - 5) - 2(s + 4) = 5 - (s + 6)$
- $5 - 2(3v - (v - 4)) = 3(v + 1) - 4(2v - 1)$
- $\frac{w + 5}{3} + \frac{w - 1}{2} = 7$
- $\frac{h - 3}{4} - \frac{h + 1}{2} = \frac{h - 9}{8}$
- $3(k - 2) + 2(2k + 1) = 7k - 4$
- $4(2j - 3) - 2(j + 5) = 6j - 20$

7. Challenge problems

- Find all real numbers x satisfying
$$3(2x - 5) - 2(3x - 7) = 5(x - 4) - (5x - 19).$$
Explain why the equation behaves differently from a typical linear equation.
- Determine all real numbers x for which

$$\frac{2x - 3}{5} - \frac{3x + 1}{7} = \frac{4x - 9}{10} - \frac{6x + 5}{14}.$$

Do not begin by clearing all denominators. Look for a structural simplification first.

- Let a be a real number. Consider the equation

$$(a + 1)((a - 1)x + 1) - 1 = (a + 1)x + 2.$$

Depending on the value of a , determine when the equation has:

- exactly one real solution;
- no solution;
- infinitely many solutions.

When the solution is unique, express x in terms of a .

8. Answer key

Practice problems

1. Let h be the number of rental hours.

$$6 + 4h < 14 + 2h$$

$$2h < 8 \implies h < 4.$$

Therefore, Company A is cheaper for

$$0 \leq h < 4.$$

2. Let m be the number of premium movies.

$$3m < 12 + m$$

$$2m < 12 \implies m < 6.$$

Since m is a whole number, Service A is cheaper for

$$m = 0, 1, 2, 3, 4, 5.$$

3. Let w be the number of weeks.

$$20 + 8w > 44 + 5w$$

$$3w > 24 \implies w > 8.$$

Therefore, Plan A contains more money

after more than 8 weeks,

or from week 9 onward.

4. Let d be the trip distance in kilometres.

$$5d < 30$$

$$d < 6.$$

Therefore, the taxi company is cheaper for

$$0 \leq d < 6 \text{ km}.$$

5. Let n be the number of notebooks.

$$4n + 10 < 6n$$

$$10 < 2n \implies n > 5.$$

Since n is a whole number, Store A is cheaper for

$$n \geq 6.$$

6. Let h be the number of parking hours.

$$3h + 5 < 3h + 9$$

$$5 < 9.$$

This statement is always true. Therefore, Garage A is cheaper for

every possible length of stay.

7. Let h be the rental duration in hours.

$$4h + 6 < 2 + 4 + 4h$$

$$4h + 6 < 4h + 6$$

$$6 < 6.$$

This statement is never true. Both shops charge exactly the same amount for every rental duration. Therefore,

Shop A is never cheaper than Shop B.

Equations

8.

$$x + 7 = 19 \implies x = 12.$$

9.

$$5y = 35 \implies y = 7.$$

10.

$$3z + 4 = 22 \implies 3z = 18 \implies z = 6.$$

11.

$$4a - 7 = 2a + 9$$

$$2a = 16 \implies a = 8.$$

12.

$$3(b + 2) = 21$$

$$b + 2 = 7 \implies b = 5.$$

13.

$$2(c + 5) + 3 = 19$$

$$2c + 13 = 19 \implies 2c = 6 \implies c = 3.$$

14.

$$3(d + 2) + d = 18$$

$$4d + 6 = 18 \implies 4d = 12 \implies d = 3.$$

15.

$$2(m + 4) + m = 2m + 17$$

$$3m + 8 = 2m + 17 \implies m = 9.$$

16.

$$3(n + 2) + n = 2(n + 5) + 8$$

$$4n + 6 = 2n + 18$$

$$2n = 12 \implies n = 6.$$

17.

$$5(p - 3) + 2p = 3(p + 1) + 10$$

$$7p - 15 = 3p + 13$$

$$4p = 28 \implies p = 7.$$

18.

$$4(2q + 1) - 3q = 2(q + 7) + 5$$

$$5q + 4 = 2q + 19$$

$$3q = 15 \implies q = 5.$$

19.

$$7 - 2(r + 4) = 3(r - 1) - 5$$

$$-2r - 1 = 3r - 8$$

$$7 = 5r \implies r = \frac{7}{5}.$$

20.

$$3(2s - 5) - 2(s + 4) = 5 - (s + 6)$$

$$4s - 23 = -s - 1$$

$$5s = 22 \implies s = \frac{22}{5}.$$

21.

$$5 - 2(3v - (v - 4)) = 3(v + 1) - 4(2v - 1)$$

$$5 - 2(2v + 4) = 3v + 3 - 8v + 4$$

$$-4v - 3 = -5v + 7 \implies v = 10.$$

22.

$$\frac{w + 5}{3} + \frac{w - 1}{2} = 7$$

Multiplying by 6,

$$2(w + 5) + 3(w - 1) = 42$$

$$5w + 7 = 42 \implies w = 7.$$

23.

$$\frac{h - 3}{4} - \frac{h + 1}{2} = \frac{h - 9}{8}$$

Multiplying by 8,

$$2(h - 3) - 4(h + 1) = h - 9$$

$$-2h - 10 = h - 9$$

$$-1 = 3h \implies h = -\frac{1}{3}.$$

24.

$$3(k - 2) + 2(2k + 1) = 7k - 4$$

$$7k - 4 = 7k - 4.$$

The equation is true for every real value of k . Therefore,

$$k \in \mathbb{R}.$$

25.

$$4(2j - 3) - 2(j + 5) = 6j - 20$$

$$6j - 22 = 6j - 20$$

$$-22 = -20.$$

This is false. Therefore,

no solution

Challenge problems – Answer key

1.

$$3(2x - 5) - 2(3x - 7) = 5(x - 4) - (5x - 19).$$

Simplifying each side,

$$6x - 15 - 6x + 14 = 5x - 20 - 5x + 19,$$

so

$$-1 = -1.$$

This identity is true for every real number. Hence,

$$x \in \mathbb{R}.$$

The equation behaves differently from a typical linear equation because all terms containing x cancel and the remaining statement is always true.

2. Let

$$A = \frac{2x - 3}{5}, \quad B = \frac{3x + 1}{7}.$$

Notice that

$$\frac{4x - 9}{10} = \frac{2x - 3}{5} - \frac{3}{10} = A - \frac{3}{10},$$

and

$$\frac{6x + 5}{14} = \frac{3x + 1}{7} + \frac{3}{14} = B + \frac{3}{14}.$$

Therefore, the equation becomes

$$A - B = A - \frac{3}{10} - \left(B + \frac{3}{14}\right).$$

Thus,

$$A - B = A - B - \frac{3}{10} - \frac{3}{14} = A - B - \frac{18}{35}.$$

Subtracting $A - B$ from both sides gives

$$0 = -\frac{18}{35},$$

which is impossible. Therefore,

no real solution.

3. Consider

$$(a+1)((a-1)x+1)-1=(a+1)x+2.$$

Expand the left-hand side:

$$(a+1)(a-1)x+(a+1)-1=(a+1)x+2,$$

so

$$(a^2-1)x+a=(a+1)x+2.$$

Move all terms to one side:

$$(a^2-a-2)x+a-2=0.$$

Factor:

$$(a-2)(a+1)x+(a-2)=0,$$

and therefore

$$(a-2)((a+1)x+1)=0.$$

We now consider the special values of a .

- If $a=2$, then

$$0=0,$$

so the equation is true for every real x . Therefore,

infinitely many solutions .

- If $a=-1$, substituting into the original equation gives

$$-1=2,$$

which is impossible. Therefore,

no solution .

- If $a \neq 2$ and $a \neq -1$, we may divide by $a-2$:

$$(a+1)x+1=0.$$

Hence

$$x=-\frac{1}{a+1}.$$

Therefore, there is exactly one real solution:

$$x=-\frac{1}{a+1}.$$

Thus,

$$\begin{cases} \text{infinitely many solutions,} & a=2, \\ \text{no solution,} & a=-1, \\ x=-\frac{1}{a+1}, & a \neq -1, 2. \end{cases}$$

9. Summarized solutions

Practice problems

1. $0 \leq h < 4$
2. $m = 0, 1, 2, 3, 4, 5$
3. $w > 8$, so Plan A has more money from week 9 onward.
4. $0 \leq d < 6$ km
5. $n \geq 6$
6.

Garage A is cheaper for every possible length of stay
7.

Shop A is never cheaper;
both shops always charge the same

Equations

8. $x = 12$
9. $y = 7$
10. $z = 6$
11. $a = 8$
12. $b = 5$
13. $c = 3$
14. $d = 3$

15. $m = 9$

16. $n = 6$

17. $p = 7$

18. $q = 5$

19. $r = \frac{7}{5}$

20. $s = \frac{22}{5}$

21. $v = 10$

22. $w = 7$

23. $h = -\frac{1}{3}$

24. $k \in \mathbb{R}$

25. No solution

Challenge problems

1. $x \in \mathbb{R}$

2. No real solution

3.

$$\begin{cases} \text{infinitely many solutions,} & a = 2, \\ \text{no solution,} & a = -1, \\ x = -\frac{1}{a+1}, & a \neq -1, 2. \end{cases}$$